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Souvik Sarkar

GitHub: SouvikSarkarJr
Portfolio: portfolio/ssjr.

Education

Master of Technology in Computer Science and Engineering **2026-2028**
Techno Main Saltlake (TMSL)

Obtained CGPA: Expected 2028

Bachelor of Technology in Computer Science and Engineering **2022-2026**
Techno India University (TIU)

Obtained CGPA: 8.19 out of 10

Publication

Self-Adaptive Improved Fungal Growth Optimization: A Hybrid Metaheuristic Framework for High-Quality, Robust and Scalable Clustering *Knowledge and Information Systems (Springer Nature)*
October 2026 [View Paper]

Project

Bi-Directional Sign Language Translator [View Project] **September 2026**
Tech Stack: Python, PyTorch, MediaPipe & Hugging Face.

- Engineered an end-to-end bi-directional translation framework converting continuous sign language into English text and vice-versa.
- Extracted high-fidelity 3D spatio-temporal hand, face, and body pose landmarks using MediaPipe to optimize feature representation and minimize computational overhead.

FGOIKMeans Hybrid Optimization [View Project] **August 2025**
Tech Stack: Python, NumPy, Pandas, Scikit-learn & Matplotlib.

- Designed and implemented a hybrid machine learning algorithm pairing **Fungal Growth Optimization (FGO)** with **K-Means** to resolve local optima stagnation.
- Achieved an **18.92% increase** in clustering quality across standard multi-dimensional benchmark datasets.
- Accelerated **convergence rates** and enhanced **model stability** by eliminating K-Means sensitivity to random centroid initialization.

Skills

Languages	Python, C, C++
AI/ML Frameworks	PyTorch, Scikit-learn, Hugging Face, SciPy, scikit-fuzzy
Data & Libraries	Pandas, NumPy, Matplotlib, GeoPandas
Web Technologies	HTML, CSS, JavaScript
Developer Tools	GitHub, VS Code, LaTeX
Documentation	Google Workspace, Microsoft Office
Game Development:	Pygame